Homework 3

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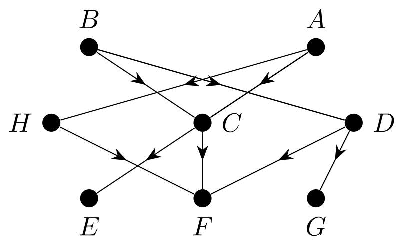
Exercise 1



$$
\begin{aligned}
A &= \{0,1\} \to 0 \\
B &= \{0,1\} \to 1 \\
C &= \max(A,B) \to 1 \\
D &= B \to 1 \\
E &= C \to 1 \\
F &= \min(H, 1 - \max(C,D)) \to 0 \\
G &= D \to 1 \\
H &= 1 - A \to 1
\end{aligned}
$$

Part (a)

Changing B potentially affects the values of C, D, F, and G. If $B = 0$, $C = \max(0,0) = 0$ and $D = 0$. Consequently, $F = \min(1, 1 - \max(0,0)) = 1$.

Part (b)

Changing B and D potentially affects the values of C, F, and G. If $B = 0$ and $D = 1$, $C = \max(0,0) = 0$. Consequently, $F = \min(1, 1 - \max(0,1)) = 0$.

Exercise 2

Part (a)

If your PhD application is a good fit for the program you apply to, you are highly likely to get admitted. However, if your application is a good fit and there are several other applicants with similar profiles, it is not highly likely that you will get admitted.

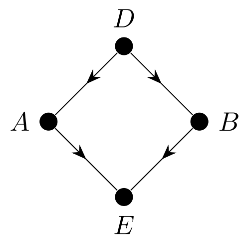
Part (b)

Let X represent the quality of your PhD application (0 if a bad fit, 1 if a good fit) and Z represent the amount of strong competition you face (0 if low, 1 if high). Letting Y represent the likelihood of you being admitted (0 if low, 1 if high), this can be modelled as $Y = \min(X, 1 - Z)$. In the original causal world, let $Z = 0$. If $X = 1$, $Y = \min(1, 1 - 0) = 1$. However, if $X = 1$ and $Z = 1$, $Y = \min(1, 1 - 1) = 0$.

Exercise 3

If $X = 0$, then $Y = 0$ and $Z = \min(0, 1 - 0) = 0$. If $X = 0$ and $Y = 0$, then $Z = \min(0, 1 - 0) = 0$. If $X = 1$, then $Y = 0$ and $Z = \min(1, 1 - 0) = 1$. If $X = 1$ and $Y = 0$, then $Z = \min(1, 1 - 0) = 0$. In both cases, the conclusion holds based on the premises. Note that it is impossible for $X = 1$ and $Y = 1$ or $X = 0$ and $Y = 1$, as that would violate the second premise. As it is always th case that the conclusion is true or one of the premises is false, the causal world fails to be a formal counterexample to cautious monotonicity.

Exercise 4



D takes on values in the range $[0, 1]$. Under the top left mapping, with weight -34.4 and bias 2.14, this is mapped to a range of $[-32.26, 2.14]$. Under the softplus function, the top variable (A) takes on values in the range $[9.780 \cdot 10^{-15}, 2.251]$. Under the top right mapping, with weight -1.30, this is mapped to a range of $[-2.927, -1.270 \cdot 10^{-14}]$. Under the bottom left mapping, with weight -2.52 and bias 1.29, this is mapped to a range of $[-1.23, 1.29]$. Under the softplus function, the bottom variable (B) taken on values in the range $[0.256, 1.533]$. Under the bottom right mapping, with weight 2.28, this is mapped to a range of $[0.585, 3.496]$. Summing and adding the final bias of -0.58, E takes on values in the range $[-2.922, 2.916]$.

$$\begin{aligned}
&D \in [0, 1] \\
&A = \ln(1 + e^{-34.4 \cdot A + 2.14}) \\
&B = \ln(1 + e^{-2.52 \cdot A + 1.29}) \\
&E = -1.30 \cdot A + 2.28 \cdot B + -0.58
\end{aligned}$$